higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## ENGINEERING PRACTICE AND MAINTENANCE
## NQF LEVEL 3

## 20 NOVEMBER 2019

**This marking guideline consists of 7 pages.**

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## QUESTION 1

| | | | | |
|---|---|---|---|---|
| 1.1 | 1.1.1 | False | | |
| | 1.1.2 | True | | |
| | 1.1.3 | True | | |
| | 1.1.4 | True | | |
| | 1.1.5 | False | | |
| | | | (5 × 1) | (5) |
| 1.2 | 1.2.1 | E | | |
| | 1.2.2 | G | | |
| | 1.2.3 | B | | |
| | 1.2.4 | F | | |
| | 1.2.5 | A | | |
| | | | (5 × 1) | (5) |
| 1.3 | 1.3.1 | B | | |
| | 1.3.2 | A | | |
| | 1.3.3 | C | | |
| | 1.3.4 | B | | |
| | 1.3.5 | B | | |
| | | | (5 × 1) | (5) |
| 1.4 | 1.4.1 | include | | |
| | 1.4.2 | Lubricant | | |
| | 1.4.3 | diagnose | | |
| | 1.4.4 | Failure | | |
| | 1.4.5 | problem | | |
| | | | (5 × 1) | (5) |
| | | | | **[20]** |

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## QUESTION 2

2.1
- Models
- Computer simulations
- Prototypes
- Pilot studies (4)

2.2 2.2.1
- Design for quality focuses on the quality of the product for the best performance.
- Design for export focuses on the international market.

2.2.2
- Benchmarking compares the performance of a product or solution against another performance to identify what needs to be improved.
- Best practices identify the best performance or product in an industry and using this as a model to guide performance.

(2 × 2) (4)

2.3
- Identifying and defining the exact problem that needs solution
- Researching various solutions to the problem
- Developing ideas to help solve the problem
- Communicating solutions and testing them out to see which ones work and which do not
- Comparing and evaluating the different solutions and designs to see how effective each one is in solving the problem
- Developing a final solution or design and then implementing it or putting it into practice (Any 4 × 1) (4)

2.4
- To use design models again and again
- To build a model of a design, without having to build an actual physical model
- To change or adjust a design model
- To develop their work on the computer screen and to save it or print it
- To show the different parts of a design and how these would be assembled or put together
- To analyse the strength and properties of different designs or parts of designs (Any 4 × 1) (4)

2.5 2.5.1 Manuals include step-by-step, practical instructions✓ on how to solve a particular engineering problem.✓

2.5.2 Workplace specifications include information and instructions✓ from the place of work about the features, use and purpose of an item.✓

(2 × 2) (4)
**[20]**

## QUESTION 3

3.1
- Direct observation
- Question the customer
- Trial and error replacement of parts
- Anticipate all the potential faults in a design
- Take action to respond to a warning sign
- Identify warning signs or symptoms that tell us something is wrong in a system or design (Any 3 × 1) (3)

3.2
- The customer's answers to your questions
- A record of your direct observation
- The data from any measurements or tests you have performed
- A diagram of the product or equipment which you used in the trial-and-error phase which tells you what is working and what is not, and therefore shows where the fault lies
- The manufacturer's servicing data and drawings, showing the different components, their maintenance and repair, and troubleshooting information
- Data and information from simulations that might have been carried out or CAD information about the operation of the equipment or system
- Servicing data or equipment history records which provide information about services, maintenance or repairs that have been carried out

(Any 4 × 1) (4)

3.3
3.3.1 Measurement-based testing
3.3.2 Operational testing
3.3.3 Direct observation technique
3.3.4 Manufacturer's service diagnostic data
3.3.5 Direct observation technique

(5 × 1) (5)

3.4 3.4.1
- Simulations can be carried out on computer✓ or in the workshop before a product is built.✓
- You can set up the system to fail so that you can see which parts are affected and which are not.✓ Often it is best to use a computer simulation when you want to set up a system to fail.✓
- In this way you can see where the fault lies and where to set the safety level to be sufficient to deal with different conditions, without putting anyone at risk. (Any 2 × 2) (4)

3.4.2
- Natural lubricants are mainly made of products from nature, such as vegetable, mineral or animal products.✓ They have a 90% petroleum or other natural base oil.
- Synthetic lubricants consist of chemicals which are artificially made and are used instead of a natural product as the base oil.✓ (2 × 1) (2)

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3.5 The manufacturer usually suggests the periods✓ when the system should be maintained to eliminate risks and dangers.✓ (2)
**[20]**

**QUESTION 4**

4.1 4.1.1 Extraction involves removing something from the earth,✓ such as coal, diamonds, iron ore, gold, platinum and oil.✓

4.1.2 A mechanical system of lubrication in an internal combustion engine✓ in which a pump forces oil onto the engine bearings.✓

4.1.3 A power plant is a processing plant,✓ where raw materials like coal are processed into energy such as electricity.✓

4.1.4 The channels of notification are the paths✓ or routes you take to inform the appropriate people about something.✓

4.1.5 Analysis of solutions is when you examine problems carefully✓ and evaluate the different aspects of it against certain criteria, looking at the strengths and weaknesses.✓

(5 × 2) (10)

4.2 Step 1: Plan and prepare for the diagnostic work.
Step 2: Verify the symptoms through observation and gathering of information.
Step 3: Define the problem area.
Step 4: Identify the possible cause of the fault.
Step 5: Determine the most probable cause and correct or repair it.

(5 × 1) (5)

4.3

| SYSTEMATIC SEARCH METHOD | RANDOM SEARCH METHOD |
|---|---|
| • Provides a process to help you investigate symptoms, locate faults and repair causes.✓ | Might not be able to tell the difference between a symptom, fault or cause.✓ |
| • Provides a process for helping you verify faults.✓ | Cannot verify the fault if you cannot tell the difference between symptoms and fault.✓ |
| • The cause is cured or treated, not the symptom. | Easy to mistake a symptom for a real fault and only repair the symptom. |
| • Efficient and uses resources and aids effectively and not wastefully. | Can spend a lot of unnecessary time, money and resources looking for faults. |
| • Can test and verify that you have repaired the fault. | Cannot verify that you have repaired the fault because you have not been through the system in a methodological way. |
| • Very useful to follow a systematic process especially with large, complex systems. | Very difficult to use this random approach when trying to locate faults in large, complex systems. |
| • Provides a step-by-step process so that you know what to do first and what to do next. | Provides no process, but uses a hit and miss method. |

(Any 2 × 2) (4)

4.4
- Cameras and monitors
- Data communications equipment
- Data converters and decoders
- Electronic track circuit equipment
- Environmental monitoring equipment
- Programmable scan tools or system testers (Any ONE) (1)

**[20]**

## QUESTION 5

5.1
- Work methods
- Lack of training
- Equipment design
- Maintenance
- Reliability (Any 3 × 1) (3)

5.2
- Machines
- Tools
- Material handling equipment (3)

5.3
- Analysis of an existing pulp and paper company's accident database
- This paper examines the number
- type and gravity of the failures involved in these events and their causes.

(Any 2 × 1) (2)

5.4
- Identify potential hazards so that you can stop accidents before they happen by doing something about these dangers.
- Remove potential hazards.
- Reduce potential hazards.
- Report potential hazards. (4)

5.5
- Temperature reduction
- Moisture displacement
- Corrosion control
- Adhesion (4)

5.6
- Hydrostatic lubrication is when oil is pumped into the bearings under pressure that is great enough to overcome the average pressure at the bearing surface.
- The parts are therefore floated on a layer of oil.
- It is mainly used in machine tools.
- Hydrodynamic lubrication can be demonstrated by the action of a journal bearing.
- Under loading the stationary shaft rests at the bottom of the bearing and there is no metal-to-metal contact.
- The majority of fluid film bearings are hydrodynamic. (Any 4 × 1) (4)

**[20]**

**TOTAL: 100**